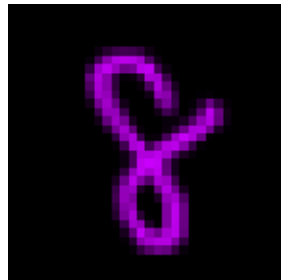
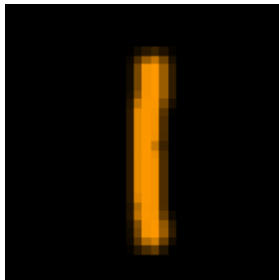
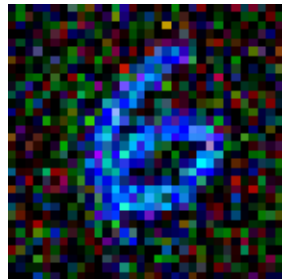
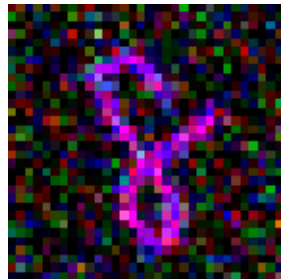
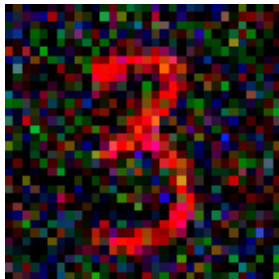
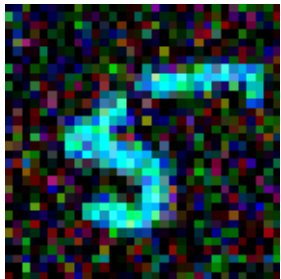
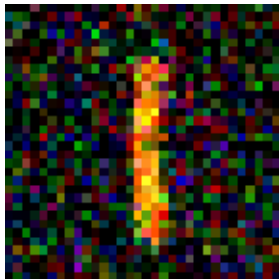


Huber Noise (alpha=1.0, scale=0.5)

Original



Huber



Camera Shifts Visualization



Huber Noise - Relative L2 (%)

Method	Alpha=0.0	Alpha=1.0
Abs-LiNGAM (Noisy)	6637.66 ± 945.80 *	7472.38 ± 948.45 *
Abs-LiNGAM (Perfect)	107569.20 ± 13304.38 *	115248.54 ± 13300.77 *
BaryCA	3724666.76 ± 425740.41 *	3770267.23 ± 428151.93 *
DiRoCA (eps_20_delta_0.0)	** 427.53 ± 224.89 **	** 505.62 ± 225.67 **
DiRoCA (eps_40_delta_0.0)	1330.93 ± 391.43 *	1604.20 ± 386.46 *
DiRoCA (eps_61.686502153383906_delta_0.0)	3672.23 ± 1004.72 *	4109.74 ± 991.20 *
DiRoCA (eps_8.0_delta_0.0)	799.27 ± 449.95	896.49 ± 449.89
GradCA	5001759.36 ± 314549.77 *	5065968.19 ± 317304.72 *

** Best Method | * Significantly worse (p<0.05) than best

Camera Shift: Rotation - Relative L2 (%)

Method		Low	Med	High
Abs-LiNGAM (Noisy)		4563.13 ± 568.96 *	4062.55 ± 547.74 *	3699.81 ± 536.39 *
Abs-LiNGAM (Perfect)		63724.99 ± 5366.03 *	50118.00 ± 4742.09 *	41143.82 ± 4236.97 *
BaryCA		2135031.34 ± 202265.62 *	1504597.62 ± 143293.06 *	1083532.37 ± 100283.63 *
DiRoCA	(eps_20_delta_0.0)	** 405.76 ± 220.54 **	** 394.39 ± 227.40 **	** 382.01 ± 231.08 **
	(eps_40_delta_0.0)	1089.40 ± 273.63 *	1008.16 ± 271.36 *	936.99 ± 263.39 *
	(eps_61.686502153383906_delta_0.0)	2669.04 ± 626.60 *	2372.56 ± 564.86 *	2125.02 ± 493.65 *
	(eps_8.0_delta_0.0)	787.12 ± 379.60	774.23 ± 361.86	773.21 ± 340.59
	GradCA	2444410.05 ± 161611.46 *	1834609.47 ± 134608.75 *	1374023.45 ± 105789.36 *

Camera Shift: Blur - Relative L2 (%)

Method		Low	Med	High
Abs-LiNGAM (Noisy)		6610.48 ± 945.59 *	6577.70 ± 945.16 *	6570.57 ± 945.03 *
Abs-LiNGAM (Perfect)		107492.64 ± 13292.10 *	107413.25 ± 13273.12 *	107397.76 ± 13268.12 *
BaryCA		3723919.71 ± 425516.59 *	3722771.51 ± 425179.67 *	3722465.66 ± 425092.15 *
DiRoCA	(eps_20_delta_0.0)	** 426.01 ± 224.29 **	** 424.31 ± 223.37 **	** 423.96 ± 223.12 **
	(eps_40_delta_0.0)	1324.21 ± 390.93 *	1316.42 ± 390.16 *	1314.78 ± 389.96 *
	(eps_61.686502153383906_delta_0.0)	3657.86 ± 1004.39 *	3639.65 ± 1003.76 *	3635.48 ± 1003.56 *
	(eps_8.0_delta_0.0)	796.99 ± 450.11	794.39 ± 450.36	793.87 ± 450.43
	GradCA	5001145.87 ± 314546.35 *	5000186.73 ± 314545.28 *	4999929.10 ± 314545.81 *

Camera Shift: Angle - Relative L2 (%)

Method		Low	Med	High
DiRoCA	Abs-LiNGAM (Noisy)	4814.74 ± 540.92 *	3011.11 ± 421.79 *	2083.06 ± 371.23 *
	Abs-LiNGAM (Perfect)	66089.81 ± 4730.41 *	26989.66 ± 1750.96 *	9205.66 ± 903.20 *
	BaryCA	3072839.80 ± 279118.05 *	1114548.85 ± 103813.18 *	278447.02 ± 21537.68 *
	DiRoCA (eps_20_delta_0.0)	** 420.23 ± 236.65 **	** 390.20 ± 212.54 **	** 275.63 ± 111.87 **
	DiRoCA (eps_40_delta_0.0)	1192.26 ± 327.35 *	908.30 ± 188.06 *	521.96 ± 81.80 *
	DiRoCA (eps_61.686502153383906_delta_0.0)	3011.02 ± 774.04 *	1971.05 ± 427.44 *	1046.55 ± 195.52 *
	DiRoCA (eps_8.0_delta_0.0)	719.77 ± 349.58	795.53 ± 347.51	803.73 ± 293.74 *
GradCA		2625469.53 ± 152059.18 *	884253.17 ± 70910.27 *	234169.05 ± 21403.54 *